

Problem Statement

September 15, 2026

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Determine the expression for the sequence term α_n when it is defined to be equal to a constant P .

Key Principles and Formulas

- **Definition of a constant sequence:** If every term of a sequence is the same fixed value P , then for all indices n we have $\alpha_n = P$.
- **Equality:** The symbol " $=$ " asserts that the two sides represent the same mathematical object.

Detailed Step-by-Step Solution

Since the definition states directly that the n -th term of the sequence equals the constant P , no further manipulation is required. Writing the definition explicitly:

$$\alpha_n = P.$$

Final Answer

$$\boxed{\alpha_n = P}$$

Study Notes and Key Takeaways

- A sequence defined by a single constant is called a *constant sequence*; all its terms are identical.
- When a problem states " $\alpha_n = P$ ", the solution is simply the given equality—no algebraic transformation is needed.
- Always verify whether additional conditions (e.g., initial values, recurrence relations) are present; if not, the definition itself is the complete answer.